

Emotion-Aware Adaptive Vehicle Control Using On-Device AI

Emotion-Aware Adaptive Vehicle Control Using On-Device AI is an intelligent driver-safety system designed to detect a driver's emotional state through their voice and respond with an appropriate vehicle safety action. Unlike conventional driver-monitoring systems that only detect or display the driver's condition, this system creates a complete “**Detect → Decide → Act**” safety loop.

The system uses a **microphone** to capture the driver's speech. The audio is processed to extract relevant voice features, which are then given to an **AI-based emotion recognition model**. The system can identify emotional states such as **calm, stress, anger, fear, and tiredness**. The AI inference is performed **locally on a Snapdragon-powered HP PC**, reducing the need to continuously send sensitive voice data to cloud servers.

Once the driver's emotional state is identified, a **safety decision module** evaluates the detected condition. If the emotional state indicates a potentially unsafe driving condition, the system generates an appropriate control command. For the prototype, this command is transmitted wirelessly to an **Arduino-based vehicle controller**, which adjusts the motor speed accordingly. For example, a high-stress or anger condition can trigger a reduction in the prototype vehicle's speed.

The key innovation of the proposed system is that **emotion detection is directly connected to vehicle control**. Instead of simply informing the driver that stress or anger has been detected, the system converts the AI prediction into a real-time safety response. This creates an adaptive vehicle system that can respond to the driver's changing emotional condition.

The use of **on-device AI on Snapdragon** also makes the system suitable for real-time edge-AI applications. The AI model can be optimized for efficient inference, allowing the system to demonstrate low-latency processing without depending on continuous cloud connectivity.

WORKFLOW

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Understand the driver's emotions | Make smart decisions | Keep the journey safe



Smarter Emotions. Safer Journeys.